Path integral formulation of quantum mechanics

$$|\psi(t)\rangle = \begin{cases} \hat{U}(t,t')\,|\psi(t')\rangle & \text{for } t>t' \\ 0 & t<t' \end{cases}$$

$$\Rightarrow \hat{U}(t,0) = \theta(t)\, e^{-i\hat{H}_0 t/\hbar} \quad , \quad \theta(t) = \begin{cases} 1, & t>0 \\ 0, & t<0 \end{cases}$$

Let $|x\rangle$ be eigenstate of position operator $\hat{x}$:

$$\hat{x}\,|x\rangle = x\,|x\rangle, \quad \langle x'|x\rangle = \delta(x'-x), \quad \int dx\, |x\rangle\langle x| = 1.$$

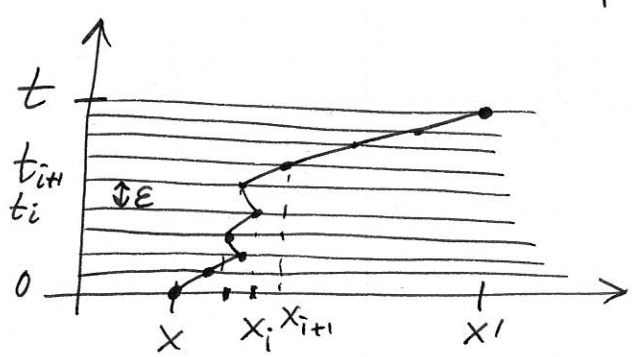
Initial state at $t=0$ is $|\psi(0)\rangle = |x\rangle$

Then $\langle x'|\psi(t)\rangle = \psi(x';t) = \langle x'|\hat{U}(t,0)|x\rangle = D(x';t;x,0)$

the propagator [from $(x,0)$ to (x',t)]

$$D(x',t;x,0) = \theta(t)\,\langle x'|e^{-i\hat{H}_0 t/\hbar}|x\rangle$$

Amplitude of propagation from x to x' in time t.



$$D(x',t;x,0) = \langle x'|e^{-\frac{i}{\hbar}\hat{H}_0(t-t_1)}\, e^{-\frac{i}{\hbar}\hat{H}_0 t_1}|x\rangle =$$

$$= \underbrace{\int dx_1}\, \langle x'|e^{-\frac{i}{\hbar}\hat{H}_0(t-t_1)}\underbrace{|x_1\rangle\langle x_1|}_{\text{complete set}} e^{-\frac{i}{\hbar}\hat{H}_0 t_1}|x\rangle =$$

$$= \int dx_1\, D(x',t;x_1,t_1)\, D(x_1,t_1;x,0)$$

Principle of amplitude superposition.

$$D(x',t;x,0) = \langle x'| \underbrace{e^{-\frac{i}{\hbar}H_0\varepsilon}\, e^{-\frac{i}{\hbar}H_0\varepsilon} \dots e^{-\frac{i}{\hbar}H_0\varepsilon}}_{N\text{ times } (\varepsilon\cdot N = t)} |x\rangle =$$

$$= \int dx_1\, dx_2 \dots dx_{N-1}\, \langle x'|e^{-\frac{i}{\hbar}H_0\varepsilon}|x_{N-1}\rangle\langle x_{N-1}|e^{-\frac{i}{\hbar}H_0\varepsilon}|x_{N-2}\rangle \dots$$

$$\dots \langle x_2|e^{-\frac{i}{\hbar}H_0\varepsilon}|x_1\rangle\langle x_1|e^{-\frac{i}{\hbar}H_0\varepsilon}|x\rangle$$

For each amplitude we can write $\left(H_0 = \frac{\hat{p}^2}{2m}\right)$

$$\langle x_{j+1}|e^{-\frac{i\varepsilon}{\hbar}\frac{\hat{p}^2}{2m}}|x_j\rangle = \int_{-\infty}^{\infty}\frac{dp_j}{2\pi}\,\langle x_{j+1}|e^{-\frac{i\varepsilon}{\hbar}\frac{\hat{p}^2}{2m}}|p_j\rangle\langle p_j|x_j\rangle =$$

Remember $\langle q|p\rangle = e^{ipq}$

$$= \int\frac{dp_j}{2\pi}\, e^{-\frac{i\varepsilon}{\hbar}\frac{p_j^2}{2m}}\, \underbrace{\langle x_{j+1}|p_j\rangle}_{e^{ip_jx_{j+1}/\hbar}}\underbrace{\langle p_j|x_j\rangle}_{e^{-ip_jx_j/\hbar}} =$$

$$= \int\frac{dp_j}{2\pi}\, e^{\frac{i}{\hbar}p_j(x_{j+1}-x_j) - i\varepsilon\frac{p^2}{2m\hbar}} =$$

$$= \int\frac{dp}{2\pi}\, e^{-\frac{i\varepsilon}{2m\hbar}\left(p^2 - \frac{2m\hbar}{\varepsilon}\frac{p(x_{j+1}-x_j)}{\hbar}\right)} = \cancel{\left(p = \sqrt{\frac{2m\hbar}{\varepsilon}}\,u\right)}$$

$$= \cancel{\frac{\sqrt{2m\hbar}}{2\pi\sqrt{\varepsilon}}}\int_{-\infty}^{\infty}\frac{dp}{2\pi}\, e^{\frac{-i\varepsilon}{2m\hbar}\left\{\left(p - \frac{m}{\varepsilon}(x_{j+1}-x_j)\right)^2 - \frac{m^2}{\varepsilon^2}(x_{j+1}-x_j)^2\right\}}$$

$$= e^{i\frac{m}{2\varepsilon\hbar}(x_{j+1}-x_j)^2}\int_{-\infty}^{\infty}\frac{dp'}{2\pi}\, e^{-i\frac{\varepsilon}{2m\hbar}(p')^2}$$

Here $p' = p - \frac{m}{\varepsilon}(x_{j+1}-x_j)$

let $p' = \sqrt{\frac{m\hbar}{\varepsilon}}\,u \Rightarrow \sqrt{\frac{m\hbar}{\varepsilon}}\int\frac{du}{2\pi}\, e^{-\frac{i}{2}u^2}$

let $I^2 = \int_{-\infty}^{\infty} \frac{du}{2\pi} \int_{-\infty}^{\infty} \frac{dv}{2\pi} \; e^{-\frac{i}{2}(u^2+v^2)} = \int_0^{\infty} \frac{2\pi}{(2\pi)^2} dr \cdot r \; e^{-\frac{i}{2} r^2} =$

$$= \frac{1}{4\pi} \int_0^{\infty} dr^2 \; e^{-\frac{i}{2} r^2}$$

$$r = r' + i r'' \Rightarrow r^2 = r'^2 - r''^2 + 2i r' r''$$

$$e^{-\frac{i}{2}(r'^2 + 2i r' r'')} = e^{-\frac{i}{2} r'^2 + r' r''}$$

The integral will converge if $r' > 0$ (as it is, $r' \in (0,\infty)$)

and $\underline{r'' < 0}$ $\Rightarrow$ assume $r'' = -0$

$$\Rightarrow I^2 = \frac{1}{4\pi} \left(\frac{2}{+i}\right) \int_0^{\infty} dz \, e^{-z} = \frac{1}{2\pi i} \Rightarrow I = \frac{1}{\sqrt{2\pi i}}$$

$$\Rightarrow \langle x_{j+1} | e^{-i\varepsilon \frac{\hat{p}^2}{2m\hbar}} | x_j \rangle = \sqrt{\frac{m\hbar}{2\pi i \varepsilon}} \; e^{i \frac{m}{2\varepsilon\hbar}(x_{j+1} - x_j)^2}$$

Thus $$D(x', t; x, 0) = \lim_{N\to\infty} \left(\frac{m\hbar}{2\pi i \varepsilon}\right)^{N/2} \int_{\{x_j\}} e^{i \frac{m}{2\varepsilon\hbar}\left\{(x'-x_{N-1})^2 + (x_{N-1}-x_{N-2})^2 + \dots + (x_1 - x)^2\right\}}$$

$$= \lim_{N\to\infty} \left(\frac{m\hbar}{2\pi i \varepsilon}\right)^{N/2} \int \prod_{j=1}^{N-1} dx_j \; e^{\frac{im}{\hbar}\varepsilon \sum_{j=1}^{N-1} \left(\frac{x_{j+1} - x_j}{\varepsilon}\right)^2}$$

$$= \int D x(t) \; e^{\frac{i}{\hbar} \int_0^t dt' \, \frac{1}{2} m (\dot{x}(t'))^2}$$

where by def $\int D\, x(t) \ldots = \lim_{N\to\infty} \left(\frac{m\hbar}{2\pi i \varepsilon}\right)^{N/2} \prod_{j=1}^{N-1} \int dx_j \ldots$

This is integral over all possible <u>paths</u>.

with boundary conditions $x(0) = x$

$x(t') = x'$

What if we have P.E. as well?

i.e. $H = \frac{\hat{p}^2}{2m} + \hat{V}(x)$

$\Rightarrow$ consider $\langle x_{j+1} | e^{-\frac{i}{\hbar}\hat{H}\varepsilon} | x_j \rangle =$

$= \langle x_{j+1} | e^{-\frac{i}{\hbar}\varepsilon\left(\frac{\hat{p}^2}{2m} + \hat{V}(x)\right)} | x_j \rangle =$

$\simeq \langle x_{j+1} | e^{-\frac{i\varepsilon}{\hbar}\frac{\hat{p}^2}{2m}} e^{-\frac{i\varepsilon}{\hbar}\hat{V}(x)} | x_j \rangle$

The error made in splitting exponential of the sum into the product is of $O(\varepsilon^2)$, which is vanishingly small in $N \to \infty$ limit.

$$e^{-i\frac{\varepsilon}{\hbar}\hat{V}(x)} | x_j \rangle = | x_j \rangle e^{-i\frac{\varepsilon}{\hbar}V(x_j)}$$

So that from each slice we pick up a factor

$e^{-\frac{i}{\hbar}\varepsilon V(x_j)}$ [can also write $V(x_{j+1})$ or $V\left(\frac{x_{j+1}+x_j}{2}\right)$]

$$\Rightarrow D(x',t;x,0) = \int Dx(t)\, e^{\frac{i}{\hbar}\int_0^t dt' \left(\frac{1}{2}m(\dot{x}(t'))^2 - V(x(t'))\right)}.$$

But $S(t) = \int_0^t dt' \left\{\frac{1}{2}m(\dot{x}(t'))^2 - V(x(t))\right\}$

is the classical action!

$\Rightarrow$ compact way to represent the amplitude of propagation from $(x,0)$ to (x',t)

$$D(x',t;x,0) = \int Dx(t)\, e^{\frac{i}{\hbar}S(t)}$$

where integral is over <u>all</u> trajectories $x(t)$.

Slightly different, but often <u>very</u> useful — the Hamiltonian path integral.

Since $\int_{-\infty}^{\infty} \frac{dp}{2\pi} e^{i\frac{p}{\hbar}x - i\frac{p^2}{2m\hbar}\varepsilon} = \sqrt{\frac{m\hbar}{2\pi i\varepsilon}}\, e^{i\frac{mx^2}{2\varepsilon}}$

we can write

$$D(x',t;x,0) = \int Dx(t)\, Dp(t)\; e^{\frac{i}{\hbar}\int_0^t dt'\left(p(t')\dot{x}(t') - H(p(t'),x(t'))\right)}$$

$$\Downarrow$$

$$\lim_{N\to\infty} \prod_{j=1}^{N-1} \int_{-\infty}^{\infty} dx_j \int_{-\infty}^{\infty} \frac{dp_j}{2\pi}$$

Note: (1) (p, x) are canonical pair: $[\hat{x}, \hat{p}] = i\hbar$

(2) $p\dot{x} - H(p,x) = L(x,\dot{x})$ where p is expressed via $x, \dot{x}$

Lagrangian

$$p = \frac{\partial L}{\partial \dot{x}}$$

Stationarity

1) $\int dt\, \delta\left(p\dot{x} - H(p,x)\right) = \int_0^t dt\left(\delta p\, \dot{x} - \delta H + p\,\delta\dot{x}\right) = \left(\delta x(0) = \delta x(t) = 0\right)$

$$= \int dt\left(\delta p\, \dot{x} - \dot{p}\,\delta x - \frac{\delta H}{\delta x}\,\delta x - \frac{\delta H}{\delta p}\,\delta p\right)$$

$$= \int dt\left\{\delta p\left(\dot{x} - \frac{\delta H}{\delta p}\right) - \delta x\left(\dot{p} + \frac{\delta H}{\delta x}\right)\right\}$$

Thus we get, requiring no linear in $\delta x, \delta p$ terms:

$$\dot{x} = \frac{\delta H}{\delta p}, \quad \dot{p} = -\frac{\delta H}{\delta x}$$

which are classical Hamilton's eqns

2) for $S = \int dt\, L(x, \dot{x})$

$$\delta S = \int dt\left(\frac{\delta L}{\delta x}\,\delta x + \frac{\delta L}{\delta \dot{x}}\,\delta\dot{x}\right) =$$

$$= \int dt\left(\frac{\delta L}{\delta x} - \frac{d}{dt}\frac{\delta L}{\delta \dot{x}}\right)\delta x$$

$\Rightarrow$ Lagrange-Euler eqn of motion: $\frac{d}{dt}\frac{\delta L}{\delta \dot{x}} - \frac{\delta L}{\delta x} = 0$

<u>Classical limit</u>:

$$D(x', t; x, 0) = \int Dx(t)\, e^{\frac{i}{\hbar} S[x(t)]}$$

For $\hbar \to 0$ slight changes from stationary path $x_{cl}(t)$ produces huge changes in $\frac{S}{\hbar}$ $\Rightarrow$ all such deviating trajectories contribute little to D — in this way we recover classical trajectories.